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Delays between Uveal Melanoma Diagnosis and Treatment Increase the Risk of Metastatic Death

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Purpose: To investigate if the interval between diagnosis and treatment of posterior uveal melanoma (UM) is associated with metastatic death.

Design: Retrospective, single-center cohort study.

Participants: A total of 1145 patients consecutively diagnosed with posterior UM at St. Erik Eye Hospital, Stockholm, Sweden, from 2012 to 2022, with recorded dates of diagnosis and primary treatment. This cohort represents 81% of all diagnosed patients in Sweden during this period.

Methods: Data on the interval between diagnosis and treatment were collected for all patients. Its prognostic importance was examined with univariate and multivariate competing risks regressions, and cumulative incidence analyses.

Main Outcome Measures: Incidence of metastatic death (UM mortality) for patients with prompt ($<$ 1 month from diagnosis) versus delayed treatment ($\geq$ 1 month) and subdistribution hazard ratios ($\exp(\beta_j)$) for every additional 10-day delay in treatment.

Results: The mean interval between diagnosis and treatment was 34 days (SD, 56, range, 0–932). Patients treated promptly had larger tumors at diagnosis, but there were no differences in patient age, tumor distance to the optic disc, rates of ciliary body involvement (CBI) or extraocular extension (EXE), or symptom duration before diagnosis. Those who were treated more than 1 month after diagnosis had greater UM mortality in American Joint Committee on Cancer (AJCC) stage II and III. In stage I, UM mortality for delayed treatment was lower for the first 10 years, followed by a marked spike in the 11th year. In multivariate competing risks regressions of all 1145 patients with tumor diameter, thickness, CBI, and EXE as covariates, the risk for UM mortality increased with 1% for every additional 10-day delay in treatment ($\exp(\beta_j)$ 1.01). Among 355 patients treated with enucleation, this delay was associated with UM mortality, independent of AJCC stage, cytomorphology, and level of immunohistochemical BAP-1 expression.

Conclusions: Increasing time between diagnosis and treatment of UM is associated with a higher risk of metastatic death. These results challenge a central concept in the understanding of metastatic progression and may indicate the existence of late metastatic seeding. They also underscore the importance of prompt treatment. Validation in independent cohorts is recommended.

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See Commentary on page 1105

The management of uveal melanoma (UM), the most common primary intraocular cancer in adults, has been an enduring challenge in ocular oncology. Central to this challenge is the unclear prognostic implication of primary tumor treatment. Although several tumor characteristics have been linked to increased metastatic risk, the impact of treatment itself is still a subject of debate.[1–4] On the one hand, successful interventions stop tumors from growing larger and seeding additional metastases—a significant concern given that larger tumors more often display aggressive gene expression profiles, loss of chromosome 3 heterozygosity, and *BAP1* mutations.[5–7] On the other hand, some experts argue that metastatic seeding occurs early in the disease, likely even before the primary tumor is diagnosed.[5,8,9] In a recent autopsy-based study, micrometastases were detected in all 5 patients who had coexisting macrometastases, as well as in 5 of the 6 patients who did not.[10] Given that 10 of the 11 patients in the study had undergone enucleation as their primary treatment—on average 11 years before their deaths—it becomes evident that the treatment occurred too late to prevent these tumor cells from migrating from the eye to establish micrometastatic niches.

Evidence from major studies, such as the Collaborative Ocular Melanoma Study, suggests that immediate treatment might not significantly influence overall survival rates for medium-sized tumors. There was no significant difference in

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overall survival between patients who received immediate treatment for medium-sized tumors and those who initially deferred treatment and opted for the Natural History Study.[11] However, at a mean of 1.4 years, 22 patients changed their minds and decided to undergo treatment after all. Furthermore, Augsburger and Vrabec[3] found no significant difference in melanoma-specific survival between patients who received prompt treatment or treatment with a median delay of 17 months. The tumors included in the latter study were small by Collaborative Ocular Melanoma Study standards, with mean diameters of 7 to 8 mm and thickness $\leq$ 3 mm, and only 30 patients were included in each group.

In a recent meta-analysis, we found that among the exceedingly rare cases of patients who deferred treatment for more than 5 years after diagnosis, 15 of 17 patients (88%) with small to medium-sized tumors developed metastases during long-term follow-up.[12] This all leads to another crucial question: If the absence of or significant delays in primary tumor treatment are associated with a great risk of metastasis, could a smaller impact be detected in a large cohort experiencing the shorter delays typical of standard clinical practice?

To address this question, the present study analyzes a large cohort of 1145 patients and examines the correlation between the interval from diagnosis to treatment and the risk of metastatic death. The findings underscore the potential prognostic importance of treatment timing in UM, challenge current prevailing theories about metastatic progression in UM, and motivate further investigations in independent cohorts.

Methods

Aim of the Study

The primary objective was to investigate whether an extended interval between the diagnosis and treatment of posterior UM increases the risk of metastatic death.

Study Design

Data were collected from consecutively diagnosed patients with posterior UM at St. Erik Eye Hospital, Stockholm, Sweden, between January 1, 2012, and December 31, 2022, having documented diagnosis and primary treatment dates (day of enucleation or insertion of ruthenium-106/iodine-125 brachytherapy plaque, n = 1173). The data included patient age and sex, largest basal tumor diameter (LBD), tumor thickness, tumor distance to the optic disc (OD), ciliary body involvement (CBI), extraocular extension (EXE, any size), prescribed radioactive dose to the tumor apex (for patients treated with brachytherapy), cytomorphology (spindle, epithelioid, or mixed, enucleations only), immunohistochemical BAP expression ($\leq$ 33% vs. $>$ 33% of tumor cell nuclei,[13] enucleations only), and length of follow-up. Patients with iris melanoma were not considered. Patient vital status (alive, died of metastatic UM, or died of other causes) had been contemporaneously registered with data provided by the National Cause of Death Registry. For a detailed account on the method for establishing the cause of death for each patient, see the Supplementary Material (available at www.aaojournal.org).

Our data for patients treated before 2012, which were recorded in another digital system, are generally less useful for determining exact dates of diagnosis and treatment. Of the initial 1173 patients, 15 were excluded because of unreliable diagnosis dates (e.g., date only derived from secondhand information, diagnosis not known at the time of treatment, diagnosis established but then questioned in discussions among faculty); 11 were excluded for having metastatic disease at presentation; and 2 were removed because of classification errors, wherein the documented treatments were not the initial interventions but were administered for recurrences after prior treatments. Consequently, 1145 patients (98% of the initial cohort) remained in the study. Because all data had already been registered in our treatment registry, no names, identification numbers, addresses, contact details, photographs, or other data that could be traced back to individual patients had to be collected. Data on histological and immunohistochemical characteristics had been registered for clinical diagnostic and prognostic purposes, and no analyses of biological tissues were performed for this project. This study was approved by the Swedish Ethical Review Authority (reference 2022-05357-01) and adhered to the tenets of the Declaration of Helsinki. Informed consent was waived because all data utilized were previously recorded in our registries, and no identifiable information such as names, identification numbers, addresses, contact details, or photographs had to be collected.

Dates of Diagnosis and Treatment

St. Erik Eye Hospital in Stockholm is the only center in Sweden where UM is diagnosed and treated with plaque brachytherapy, and the only center with a dedicated ophthalmic pathology laboratory that examines the eyes of patients who have been recommended enucleation. Although ophthalmologists from other Swedish clinics might suspect a choroidal tumor and refer patients to our facility, a definitive UM diagnosis is made only after examination at our center because of specific equipment and specialized expertise requirements. We encourage other health care professionals, such as optometrists and nurses, to direct patients to general ophthalmologists for preliminary assessments before referral to us. Some patients with choroidal nevi or indeterminate melanocytic lesions are monitored until markers such as growth, orange pigment, subretinal fluid, or ultrasound hollowness confirm UM.

For the purpose of this study, the date of the UM diagnosis—and not the date of discovery of any precursor lesion—is used to determine the time span between diagnosis and treatment. Consequently, the duration a lesion was observed before a UM diagnosis does not impact the calculated delay times.

At our facility, we review each patient's medical history, prior diagnoses, medication regimen, and past ocular examinations. Diagnostic procedures include testing for best-corrected visual acuity (BCVA) and intraocular pressures, wide or standard field fundus photos with autofluorescence, OCT for discernible tumors, slit-lamp biomicroscopy, and A- and B-scan ultrasonography. After this assessment, a diagnosis of UM can be established in nearly all affected patients. In rare instances where clinical examinations are inconclusive, we undertake either transvitreal or transscleral biopsies.[14,15] The day and time of every UM diagnosis are logged in our treatment registry, which was validated against patient medical records for the present study.

Treatment recommendations are typically made on the day of diagnosis. Modalities, determined largely by tumor size, location, and patient preference, include the following:

- Tumors $\leq$ 16 mm in diameter and $\leq$ 6 mm thick: plaque brachytherapy with ruthenium-106
- Tumors $\leq$ 16 mm in diameter and 6- to 10-mm thick: plaque brachytherapy with iodine-125
- Larger tumors: enucleation

The prescription point for plaque brachytherapy was the tumor apex. For ruthenium-106 and iodine-125, 100 and 80 gray (Gy) were prescribed, respectively. No minimum scleral dose was used.

Extrascleral extension was an indication for enucleation, unless minimal. The surgical procedures and follow-up after treatment have been described in detail.[16]

For patients advised to undergo enucleation, the procedure can be performed at their local clinics. Nevertheless, enucleated eyes are forwarded to us. This enables us not only to collect the surgery date for patients treated elsewhere but also to identify histopathological and immunohistochemical characteristics. When evaluating BAP-1 immunoreactivity, a strong marker for poor prognosis, the mean score from the 3 high-power fields was used for classification.[17,18] Tumors with positive staining in $> 33\%$ of cell nuclei were classified as having a high/retained expression and tumors with positive staining in $\leq 33\%$ of cell nuclei as having low/loss of expression, as described previously.[19] The staining protocol and BAP-1 scoring procedure are described in further detail in the Supplementary Material (available at www.aaojournal.org).

Factors Influencing the Timing of Uveal Melanoma Treatment

The interval between a diagnosis of UM and its treatment is influenced by several factors. Peaks in diagnoses during busy periods can lead to waiting times of several weeks for an available radioactive plaque or operating room. This wait may be extended during holiday seasons when staff and facility availability is reduced. Some patients, upon diagnosis, may be hesitant about treatment and require additional time for consideration and family discussions. Yet, consultations among our faculty concerning the definitive diagnosis seldom cause delays, given our daily interactions within our specialized group of ocular oncologists.

Tumor-related characteristics rarely exert a substantial impact on the timing of treatment. If anything, treatment may be perceived to be more urgent for patients with very large tumors, whereas instances of delays may arise from uncertainties on the part of physicians or patients, particularly when dealing with very small tumors. In this study, patient- and tumor-related differences between those who receive prompt treatment and those subject to treatment delays will be examined more closely.

Likewise, the patient's place of residence or socioeconomic status has negligible influence on treatment timing. Given Sweden's tax-funded health care system, patients incur a nominal fee per appointment, capped annually, regardless of the treatment or examination type. Upon diagnosis and treatment determination, typically during the same visit, the coordinating nurse is tasked with scheduling the surgical procedure at the earliest available slot. This is done according to the same checklist irrespective of the patient's residence, ensuring facilities and requisite personnel, including medical physicists from Karolinska University Hospital, are available. Patients receive notifications via mail about their scheduled surgery 10 to 14 days in advance. For those residing outside Stockholm, transportation to St. Erik Eye Hospital is arranged. Patients diagnosed in our clinic but undergoing enucleation in their home region are usually treated by an experienced oculoplastic surgeon within the same timeframe. The interval between diagnosis and treatment will not influence who performs the treatment, for example, patients will not be managed by a less experienced team if there is a delay between diagnosis and treatment.

The most significant factor leading to treatment delays, other than the ones mentioned, is the presence of urgent comorbidities. Patients with conditions such as cardiac arrhythmias or severe infections might face treatment postponements. These conditions may be associated with mortality from unrelated causes, but they are not known to increase the risk of death due to metastatic UM. Obesity seems to be associated with a lower risk for metastatic death, but rarely influences the timing of plaque brachytherapy or enucleation.[20] In this study, any patient experiencing a delay exceeding 100 days will undergo a detailed examination to ascertain the specific reason for the gap between diagnosis and treatment.

Power Calculation

A post hoc power calculation was conducted. With a sample size of 1145 patients, a power of 80%, and an α of 0.05, this study was designed to detect differences in mortality of approximately 8 percentage points, assuming a 12-year incidence of metastatic death of 40% for the group with highest incidence.[21]

Statistical Analyses

Differences with a $P < 0.05$ were considered significant, and all P values were 2-sided. The Shapiro−Wilk test was used to evaluate the deviation of continuous variables from a normal distribution. If the test was significant ($P < 0.05$), the Mann−Whitney U test was used to compare groups; otherwise, the Student t test was used. Pearson's chi-square test (if all fields had a sample of ≥ 5) or Fisher exact test (if any field had a sample of < 5) was used in contingency tables. We used 2 complementary methods to determine whether our follow-up data for patients treated within 1 month from diagnosis (prompt) or later (delayed) met the proportional hazard assumption: First, we built a Cox regression model to calculate the hazard ratio for UM-related and all-cause mortality with a time-dependent (the time variable $T\times$[delay vs. prompt]) versus a time-independent variable (delay vs. prompt) as covariates. Second, we used a graphical approach and assessed log-minus-log-transformed survival curves. If P was > 0.05 in Cox regression with a time-dependent versus a time-independent variable, and if survival curves were parallel without any crossing or divergence, the proportional hazard assumption was considered fulfilled. Cumulative competing risks incidence function estimates for those with prompt versus delayed treatment were plotted with the cmprsk package for R (R Core Team).[22] The equality of survival distributions was tested with Gray's test for equality. Deviations in radioactive dose to tumor apex from the prescribed dose due to changes in plaque brachytherapy treatment duration (e.g., plaque removal 30 minutes later than planned due to operating room delays) were calculated according to a previously used equation:[23,24]

$$\left(\frac{actual\ brachytherapy\ duration\ (hours)}{planned\ brachytherapy\ duration\ (hours)}\right) * prescribed\ dose = estimated\ actual\ dose$$

To estimate the change in tumor volume between diagnosis and treatment, tumors were modeled as semi-ellipsoids, a shape previously associated with prognosis:[7]

$$\text{Estimated volume of tumor} = \frac{\pi}{6} \times thickness \times \text{LBD} \times (\text{LBD} \times 0.85)$$

Univariate and multivariate competing risks regressions for patients treated after increasing 10-day intervals after diagnosis; with increasing LBD; thickness; CBI versus no CBI; EXE versus no EXE; with epithelioid versus mixed versus spindle tumor cells (enucleations only); and with loss of BAP-1 expression versus retained BAP-1 expression (enucleations only) were performed with the crr package for R.[25] All other statistical analyses were performed using IBM SPSS statistics version 27 and GraphPad Prism version 10.0.2.

Results

Descriptive Statistics

Our study sample of 1145 patients represents 81% of the 1413 posterior UM diagnoses recorded in Sweden from January 1, 2012, to December 31, 2022. The discrepancy of 268 patients (19%) arose from missing data regarding the date of diagnosis, treatment, or both in the treatment registry and medical records. Of these 1145 patients, 554 (48%) were female and 591 (52%) were male. The mean LBD and thickness were 10.5 mm (standard deviation [SD], 3.6) and 5.1 mm (SD, 2.6), respectively. Patients treated within 1 month from diagnosis had greater LBD (11.7 [SD, 4.0] vs. 11.1 mm [SD, 4.4], Mann−Whitney U $P = 0.02$) and tumor thickness (5.9 [SD, 3.5] vs. 5.3 mm [SD, 3.4], Mann−Whitney U $P < 0.001$); had more advanced American Joint Committee on Cancer (AJCC) T-category (chi-square $P = 0.01$) and AJCC stage (chi-square $P = 0.007$); and were more likely to have been treated with enucleation (34% vs. 26%, chi-square $P = 0.003$). There was no difference in patient age at diagnosis, tumor distance to the OD, or rates of CBI or EXE (Table 1). Table S2 (available at www.aaojournal.org) presents the median interval between diagnosis and treatment for patients treated within 1 month of diagnosis or later, categorized by AJCC stage.

The mean interval between diagnosis and treatment was 34 days (SD, 56, range, 0−932). Notably, 17 patients experienced delays exceeding 100 days before treatment. Of these, 7 initially declined surgery but later consented to the recommended treatment. Six patients, diagnosed with small melanomas in the posterior pole, faced treatment delays due to hesitancy from either the attending ophthalmologist or the patients themselves, stemming from concerns over potential severe vision impairment. The remaining 4 patients had to address significant comorbidities, such as malignant hypertension, high-gradient aortic stenosis, and kidney failure, before undergoing general anesthesia. Ultrasonography and wide-field fundus photography were used to remeasure the tumors of 12 of the 17 patients just before their delayed treatments. After a mean delay of 390 days, the mean primary tumor volume had increased from 355 to 769 mm^3, corresponding to an annual increase of 109% (Wilcoxon matched-pairs signed-rank test, $P < 0.001$, Fig S1, available at www.aaojournal.org). Of the 17 patients, 8 eventually died of metastatic UM, and 2 died of unrelated causes at a median of 5.1 years after UM diagnosis. At the end of the follow-up period, 7 of the 17 patients were still alive at the end of follow-up at a median of 7.1 years postdiagnosis.

A total of 209 patients (18%) had undergone examinations for a choroidal nevus or indeterminate melanocytic lesion before being diagnosed with UM. However, the duration of these observations was often inconsistently recorded. For example, specifics were missing in referrals, subsequent exams omitted initial findings, or records existed before a transition to a digital system. Consequently, these duration data were deemed unreliable and not analyzed further. The number of patients observed before UM diagnosis was similar among those who received prompt versus delayed treatment (chi-square $P = 0.51$). Patients observed before UM diagnosis had tumors with significantly smaller diameters (median 7.0 vs. 12.0 mm, Mann−Whitney U $P < 0.001$) and thickness (median 2.5 vs. 5.5 mm, Mann−Whitney U $P < 0.001$) than those who were diagnosed without previous observation.

Data on self-reported symptom duration (e.g., floaters, shadow in the visual field, blurry vision) before diagnosis were available for 415 patients. No significant difference was observed in the duration of symptoms between patients treated within 1 month from diagnosis and those treated after 1 month (median: 2 months in both groups, Mann−Whitney U $P = 0.38$). Patient follow-up, age, sex, and symptom duration characteristics are further illustrated in Figure 2.

Median follow-up times were as follows: (a) 4.1 years (range, 0.4−10.5 years; interquartile range [IQR], 4.8 years) for all 1145 patients; (b) 2.7 years (range, 0.4−10.5 years; IQR, 2.4 years) for 194 patients who died of metastatic UM; (c) 3.7 years (range, 0.0−10.3 years; IQR, 3.5 years) for 127 patients who died of other causes; and (d) 4.6 years (range, 0.5−10.5 years, IQR, 5.2 years) for 824 survivors. Our follow-up data for time to metastatic death for patients with prompt versus delayed treatment met the proportional hazards assumption in a Cox regression with a time-dependent versus a time-independent variable ($P = 0.90$). The visual assessment of log-minus-log-transformed survival curves agreed (Fig S3, available at www.aaojournal.org).

Distribution of Other Factors

In addition to the patient, tumor, and treatment factors reported above, we examined the distribution of potential confounders in the subsets with available data.

There were no significant differences in logarithm of the minimum angle of resolution BCVA; deviations from planned plaque brachytherapy treatment duration; cytomorphology (spindle vs. mixed vs. epithelioid); or in the presence of vasculogenic mimicry (extracellular networks, closed loops, arcs with branching, or any combination of these) between those who received prompt versus delayed treatment (Fig 4A−D). A significantly higher proportion of patients treated promptly had loss of immunohistochemical BAP-1 expression in their tumor cell nuclei (chi-square $P = 0.009$, Fig 4E).

Patients' self-reported symptoms at the time of UM diagnosis were similar between those who received prompt versus delayed treatment (Fig 4F−J).

Forty-five of the 790 patients treated with plaque brachytherapy experienced local tumor recurrence. The cumulative incidence of local recurrences was not significantly different between the prompt and delayed treatment groups, with rates of 7% and 5% at 12 years, respectively (Fig S5, available at www.aaojournal.org).

Incidence of Metastatic Death

For the 394 patients in AJCC stage I, our cumulative incidence analysis, derived from competing risks data, revealed no significant difference in the incidence of death due to metastatic UM (UM mortality) between patients who received prompt treatment (within 1 month from diagnosis) and those who received delayed treatment ($\geq$ 1 month from diagnosis). Interestingly, the incidence curve for delayed treatment displayed a marked spike in UM mortality 10 to 11 years postdiagnosis (Fig 6A).

In contrast, the 593 patients with delayed treatment in AJCC stage IIA+IIB demonstrated a significantly increased UM mortality (28% vs. 23%, Fig 6B). This trend was consistent for the 158 patients in AJCC stages IIIA+IIIB+IIIC (79% vs. 49%, Fig 6C), as well as when considering all 1145 patients irrespective of stage (34% vs. 23%, Fig 6D).

As for mortality from causes other than metastatic UM, a significant difference was observed only in AJCC stage I (31% vs. 14% for delayed vs. prompt treatment). Incidence curves for UM mortality and other causes for each treatment modality (plaque brachytherapy with ruthenium-106, iodine-125, or enucleation) are shown in Figure S7 (available at www.aaojournal.org).

Competing Risks Regressions

In univariate competing risks regressions of data from all 1145 patients, time between UM diagnosis and treatment (per additional 10-day interval); LBD; tumor thickness; CBI; and EXE were all associated with UM mortality.

In multivariate analysis, time between UM diagnosis and treatment; LBD; and tumor thickness retained their significance.

Table 1. Characteristics of Included Patients, Tumors, and Treatment Variables

	Prompt Treatment,* n = 713	Delayed Treatment,† n = 432	P
Sex, n (%)			0.91#
Female	344 (48)	210 (49)	
Male	369 (52)	222 (51)	
Age at UM diagnosis, mean yrs (SD)	65 (14)	66 (13)	0.63**
Tumor LBD, mean mm (SD)	11.7 (4.0)	11.1 (4.4)	0.02**
Tumor thickness, mean mm (SD)	5.9 (3.5)	5.3 (3.4)	< 0.001**
Tumor distance to OD, mean mm (SD)	3.8 (3.3)	3.9 (3.8)	0.47**
CBI, n (%)	71 (10)	47 (10)	0.62#
EXE, n (%)	40 (6)	27 (6)	0.66#
Observation period before UM diagnosis,‡ n (%)			0.51#
No	587 (82)	349 (81)	
Yes	126 (18)	83 (19)	
AJCC T-category at diagnosis, n (%)			0.01#
T1a	216 (30)	178 (41)	
T1b	7 (1)	8 (2)	
T1c	11 (2)	2 (1)	
T1d	4 (1)	1 (< 1)	
T2a	222 (31)	106 (25)	
T2b	10 (1)	9 (2)	
T2c	3 (< 1)	3 (1)	
T2d	1 (< 1)	4 (1)	
T3a	143 (20)	70 (16)	
T3b	28 (4)	11 (3)	
T3c	10 (1)	6 (1)	
T3d	2 (< 1)	2 (1)	
T4a	31 (4)	14 (3)	
T4b	16 (2)	9 (2)	
T4c	5 (1)	6 (1)	
T4d	2 (< 1)	3 (1)	
T4e	2 (< 1)	0 (0)	
AJCC stage at diagnosis, n (%)			0.007#
I	216 (30)	178 (41)	
IIA	244 (34)	117 (27)	
IIB	153 (22)	79 (18)	
IIIA	73 (10)	38 (9)	
IIIB	23 (3)	17 (4)	
IIIC	4 (1)	3 (1)	
Treatment			
Ru-106 plaque brachytherapy, n (%)	322 (45)	207 (48)	0.003#
I-125 plaque brachytherapy, n (%)	146 (21)	115 (27)	
Enucleation, n (%)	245 (34)	110 (26)	
BAP-1 IHC§			0.009#
High, n (%)	20 (38)	17 (71)	
Low, n (%)	32 (62)	7 (29)	
Radioactive dose to tumor apex,‖ median Gy (IQR)	100 (20)	100 (20)	0.08**
Adjunct TTT,¶ n (%)	83 (18)	51 (16)	0.49#
Median follow-up yrs (IQR)			
Died of metastatic UM	3.2 (3.0)	2.4 (2.0)	
Died of any other cause	3.9 (3.6)	3.0 (3.4)	
Survivors	5.6 (5.2)	3.7 (3.1)	

AJCC = American Joint Committee on Cancer; CBI = ciliary body involvement; EXE = extraocular extension; Gy = gray; IHC = immunohistochemical; IQR = interquartile range; I-125 = iodine-125; LBD = largest basal diameter; OD = optic disc; Ru-125 = ruthenium-125; SD = standard deviation; TTT = transpupillary thermotherapy; UM = uveal melanoma.
*Treated within 1 month from UM diagnosis.
†Treated more than 1 month after UM diagnosis.
‡Patient observed for a choroidal nevus or indeterminate melanocytic lesion before a UM diagnosis was established.
§For 76 enucleated patients with available data. High: immunohistochemical BAP-1 expression in >33% of tumor cell nuclei. Low: immunohistochemical BAP-1 expression in ≤33% of tumor cell nuclei.
‖For 790 patients treated with ruthenium-106 or iodine-125 plaque brachytherapy.
¶TTT to central tumor margins (i.e., closest to the OD and fovea) at the time of primary brachytherapy.
#Chi-square test.
**Mann−Whitney *U* test.

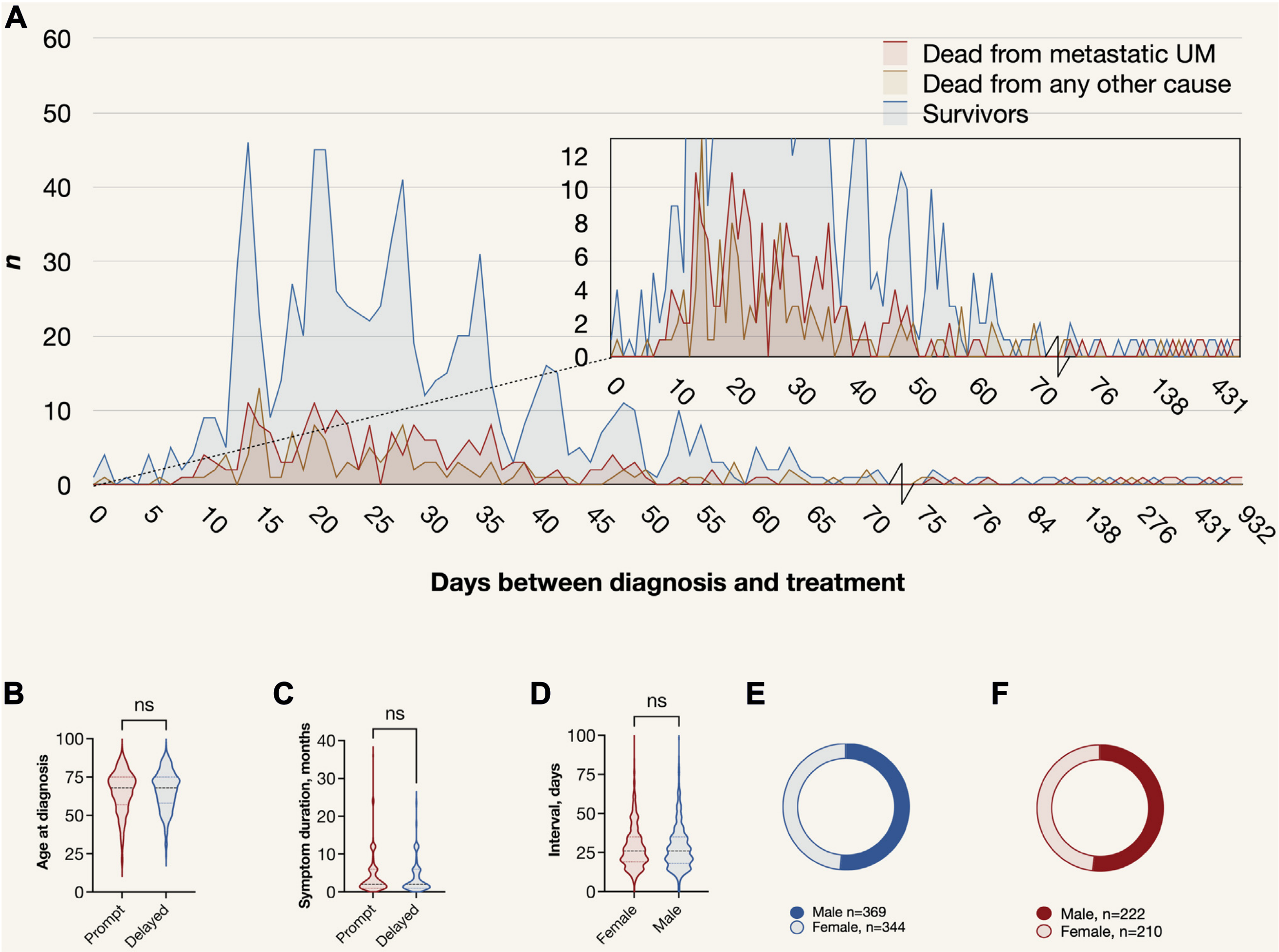

Figure 2. Patient age, sex, and symptom duration characteristics versus treatment delay. **A,** Number of days between posterior uveal melanoma (UM) diagnosis and treatment for n patients by vital status. Insert: Magnified y-scale. There were no significant differences in **(B)** patient age at diagnosis or **(C)** self-reported symptom duration in months before diagnosis between those with prompt (within 1 month from diagnosis) or delayed ($\geq$ 1 month from diagnosis) treatment (Mann–Whitney *U* $P = 0.63$ and 0.38, respectively). **D,** The interval between diagnosis and treatment was similar for female and male patients (median: 26 days in both groups, $P = 0.75$), and male and female patients were similarly distributed among those with **E**, prompt versus **F**, delayed treatment (chi-square $P = 0.91$). ns = not significant.

For every additional 10-day delay, the risk for UM mortality increased with 1% (subdistribution hazard ratio [exp(β_j)] 1.01, 95% confidence interval [CI], 1.00–1.03, $P = 0.02$).

When analyzing patients treated with plaque brachytherapy separately, time between UM diagnosis and treatment was not associated with UM mortality in univariate analysis (delays were more likely for small tumors) but was a significant predictor in multivariate analysis with LBD and tumor thickness as covariates (exp(β_j) 1.02, 95% CI, 1.01–1.03, $P < 0.001$).

For enucleated patients, time between UM diagnosis and treatment was associated with UM mortality (exp(β_j) 1.07, 95% CI, 1.04–1.10, $P < 0.001$) independent of AJCC stage, cytomorphology, and immunohistochemical BAP-1 expression (Table 3).

Discussion

Main Findings

In this study, we demonstrate that increasing time between diagnosis and treatment of posterior UM is associated with increased risk for metastatic death. This association was consistent for those treated with plaque brachytherapy, adjusted for tumor diameter and thickness; for those treated with enucleation, adjusted for AJCC stage, cytomorphology, and BAP-1 expression; and for all patients combined, adjusted for tumor diameter, thickness, CBI, and EXE. There were no differences in patient age at diagnosis, tumor distance to the OD, BCVA, presence of vasculogenic mimicry, deviations from planned plaque brachytherapy treatment duration, or self-reported symptoms between the groups—of which some have prognostic importance.[26–28]

In AJCC stage II and III, patients treated $\geq$ 1 month from diagnosis had greater UM mortality, but not in stage I. This is surprising, because the prevailing theory postulates that the metastases that lead to death for patients with large tumors have left the eye and colonized distant organs, primarily the liver, several years before diagnosis.[5,8,9]

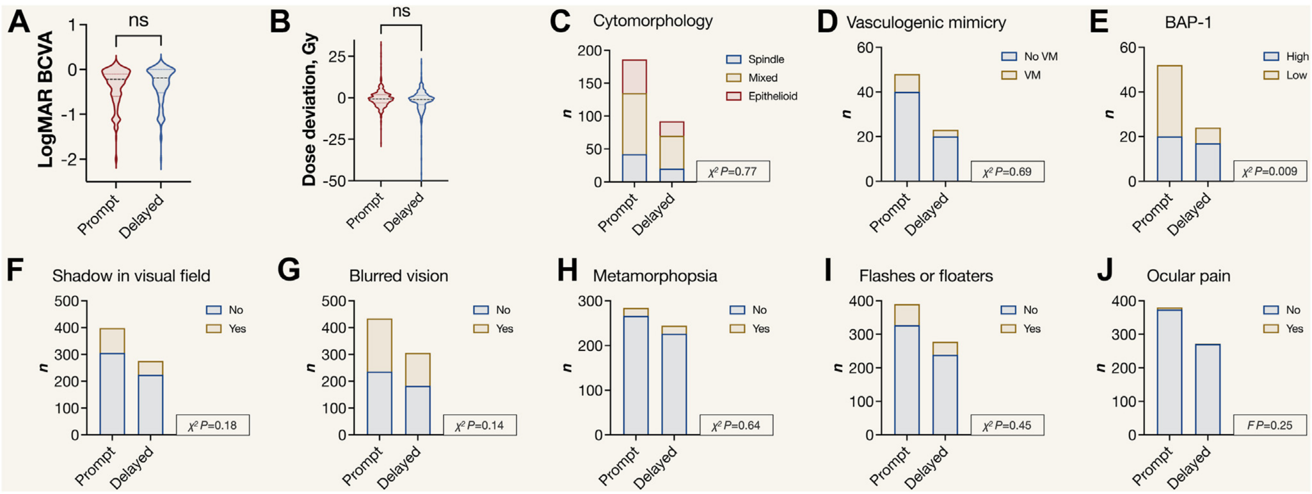

Figure 4. Further characteristics of patients with prompt (within 1 month from diagnosis) versus delayed ($\geq$ 1 month from diagnosis) treatment. There were no differences in **(A)** best-corrected visual acuity (BCVA) at the time of uveal melanoma (UM) diagnosis (data available for n = 835, median logarithm of the minimum angle of resolution −0.22 vs. −0.19, Mann−Whitney *U* *P* = 0.17); **(B)** deviation in radioactive dose to tumor apex from the prescribed dose due to changes in plaque brachytherapy treatment duration (e.g., plaque removal 30 minutes later than planned due to operating room delays, n = 784, median difference −0.7 vs. −1.0 Gy, Mann−Whitney *U* *P* = 0.11); **(C)** cytomorphology (spindle vs. mixed vs. epithelioid, n = 278, chi-square *P* = 0.77); or **(D)** presence of vasculogenic mimicry (extracellular networks, closed loops, arcs with branching, or any combination of these, n = 71, chi-square *P* = 0.69). **E,** Patients with prompt treatment were more likely to have low immunohistochemical expression of BAP-1 in tumor cell nuclei (< 33%, n = 76, chi-square *P* = 0.009. Patients' self-reported symptoms at the time of UM diagnosis were similar between those who received prompt versus delayed treatment. Specifically, there were no significant differences in **(F)** a shadow in the visual field (n = 674, chi-square *P* = 0.18), **(G)** blurred vision (n = 740, chi-square *P* = 0.14), **(H)** metamorphopsia (n = 528, chi-square *P* = 0.64), **(I)** flashes or floaters (n = 668, chi-square *P* = 0.45), or **(J)** ocular pain (n = 652, Fisher exact *P* = 0.25). Gy = gray; logMAR = logarithm of the minimum angle of resolution; ns = not significant.

Revisiting Prevailing Theories

A survival effect of delayed treatment has been more expected for small tumors.[4] The late spike in UM mortality seen in AJCC stage I could be a sign of this effect. It suggests that metastases could have spread from small tumors during the gap between diagnosis and treatment and remained dormant for a decade before manifesting clinically. However, our data also indicate that metastases were formed by cells that left larger primary tumors during this gap. Consequently, our results challenge a central concept in the understanding of the metastatic progression in UM. To explain the strong prognostic implication seen for large tumors treated with a delay herein, metastatic tumor cells that leave the eye in more advanced stages of the disease had to grow relatively fast, perhaps without the initial period of dormancy that has become a hallmark of this disease.

This could corroborate the prognostic implication of local tumor recurrences: Vrabec et al[29] and Gragoudas et al[30] have reported significantly shorter survival among patients who experience local tumor recurrences after eye-preserving therapy. In 2016, an international collaboration reported that a group of patients who had local tumor recurrence had shorter metastasis-free survival.[31] Nevertheless, it has been debated whether the tumor recurrence is the cause of metastatic disease in itself or merely an indicator of increased metastatic potential from the onset.[31] The latter has historically been favored by most experts, but perhaps the balance is shifting: Recently, we showed that there is a higher rate of local recurrences after plaque brachytherapy with 15- versus 20-mm ruthenium-106 plaques if the position is not verified after scleral attachment.[16] Patients who had local recurrences had higher UM mortality, even though they had significantly smaller tumors. This view was corroborated in a recent study by Bagger et al,[32] who found that tumors that recur after plaque brachytherapy are not more likely to have abnormal chromosome 3 and 8q status. Recently, Sreenivasa and colleagues[33] reported that the time between UM diagnosis and treatment with linear accelerator−based fractionated stereotactic radiosurgery was associated with reduced local recurrence-free and enucleation-free survival.

Implications for Understanding Metastatic Progression

As underscored by Damato,[1] it remains uncertain how many tumors, initially lacking high-risk features such as monosomy 3 and 8q gain, will later manifest these aggressive markers if untreated. However, estimations of early metastatic seeding must not be interpreted as indicative that all tumors have aggressive traits from the onset or even that all the tumors that develop such traits do so when they are very small. Likewise, the concept of early seeding does not mean that it occurs once and then never again. Evidently, one might propose that it occurs continuously until the primary tumor has been effectively treated and that metastases that leave the eye in later stages of the disease are at a higher risk of establishing growing,

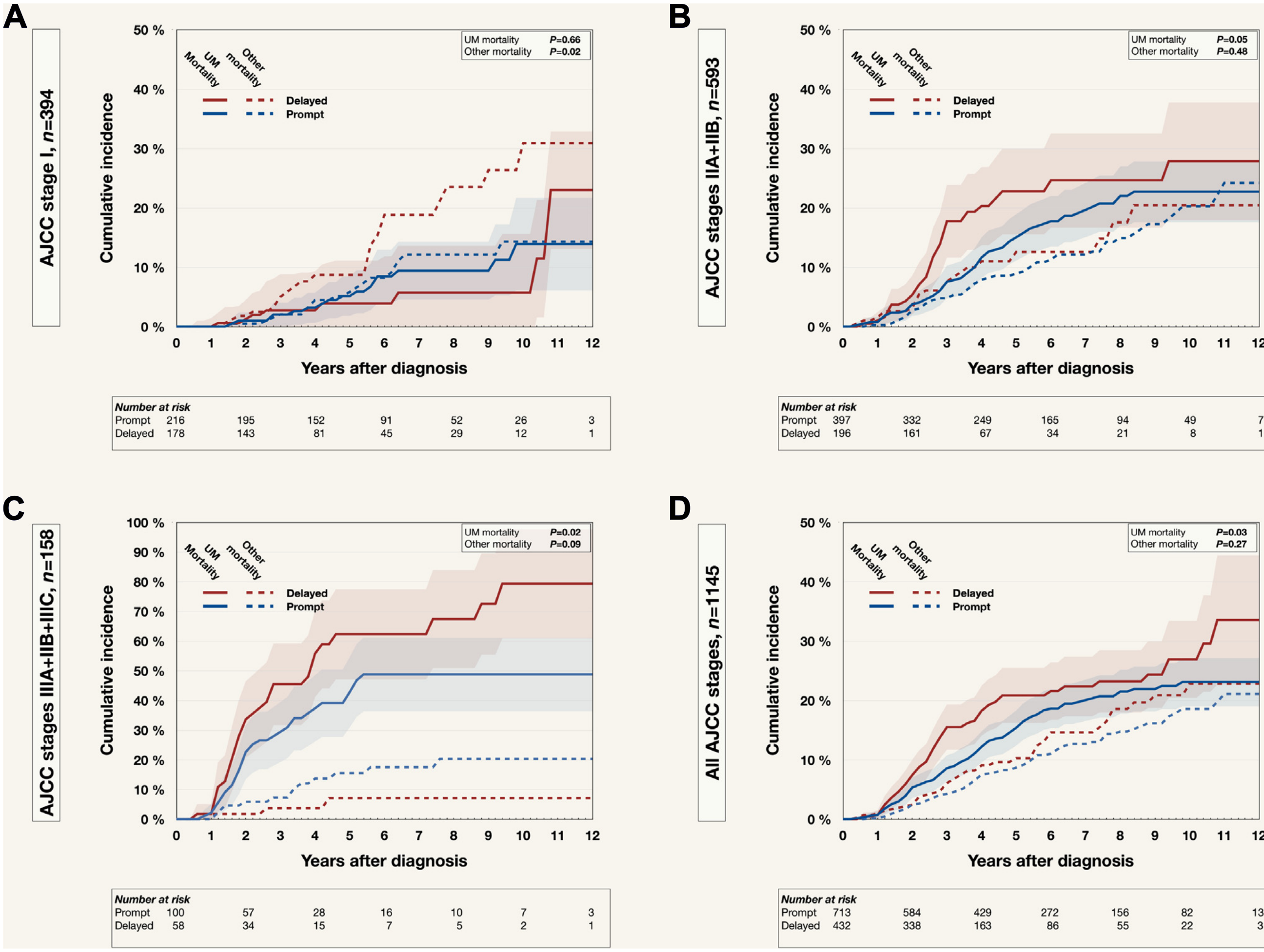

Figure 6. Cumulative incidence of death from competing risks data for **(A)** 394 patients in American Joint Committee on Cancer (AJCC) stage I; **(B)** 593 patients in AJCC stage IIA or IIB; **(C)** 158 patients in AJCC stage IIIA, IIIB, or IIIC; and **(D)** all 1145 patients. Colored fields indicate 95% confidence intervals (CIs). Note that Y-scale varies. Prompt: treated within 1 month from diagnosis. Delayed: treated ≥ 1 month from diagnosis. UM = uveal melanoma.

fatal macrometastases. There exists a spectrum of tumor behavior, with some tumors presenting as small and indolent and unlikely to express other aggressive features upon initial observation. These tumors may later undergo a transformation, acquiring malignant traits over time.[34,35] Regardless of the size of this subgroup, early treatment would likely remove their contribution to metastatic rates. This principle is exemplified by the prognosis for iris melanoma, typically treated at a much smaller size and thereby associated with a lower risk of metastasis. Yet, when adjusted for size, metastasis-free survival rates between iris and choroidal melanomas are comparable.[36] Observation of tumor growth often serves as the rationale for categorizing a pigmented choroidal mass as melanoma. Previous research indicates that more than one-third of suspected and more than half of diagnosed small UMs will demonstrate growth. Moreover, the tumors that do grow exhibit a substantial risk of metastasis—exceeding 40% 10 years post-treatment.[37–39] Thus, there can be little room for doubt: A portion of tumors that do not initially have aggressive traits will develop those if they are left untreated or if they recur after primary treatment. These tumors will then be at increased risk for metastasis. If tumors are treated before such aggressive traits are developed, even if only in a small portion of tumors, survival should be improved. As in any cancer, early diagnosis and treatment must be considered of utmost importance for patients with UM.

Strengths and Limitations

This study expands on a central question in ocular oncology, which has been examined in smaller cohorts, of smaller tumors, with shorter follow-up.[3] Being the only center in Sweden to diagnose UM, to provide plaque brachytherapy, to have a dedicated ophthalmic pathology laboratory that examines all enucleated eyes, and to have meticulous control over the follow-up of all patients, we have a uniquely suitable setting for this type of study. Further, we examined the association between delays in treatment and UM mortality with competing risks data rather than Kaplan–Meier curves. In the presence of competing risks (i.e., death due to causes other than metastatic melanoma), the former is preferrable.

Table 3. Univariate and Multivariate Competing Risk Regressions, and Subdistribution Hazard Ratio ($\exp(\beta_j)$) for Death Due to Metastatic Uveal Melanoma

		β_j	SE	Z-score	P	$\exp(\beta_j)$	95% CI	Wald
All, n = 1145	**Univariate**							
	Interval*	0.02	0.01	4.4	<0.001	1.02	1.01–1.03	19.0
	LBD†	0.24	0.02	12.8	<0.001	1.27	1.22–1.31	164.0
	Thickness†	0.19	0.02	9.4	<0.001	1.20	1.16–1.25	87.6
	CBI‡	0.85	0.20	4.4	<0.001	2.34	1.59–3.42	18.9
	EXE§	1.16	0.23	5.0	<0.001	3.19	2.02–5.01	25.0
	Multivariate							
	Interval*	0.01	0.01	2.3	0.02	1.01	1.00–1.03	5.2
	LBD†	0.20	0.02	9.4	<0.001	1.22	1.17–1.27	88.1
	Thickness†	0.05	0.02	2.0	0.05	1.05	1.00–1.10	3.9
	CBI‡	0.27	0.20	1.3	0.18	1.31	0.88–1.96	1.8
	EXE§	0.37	0.30	1.2	0.21	1.45	0.81–2.60	1.5
Plaque brachytherapy, n = 790	**Univariate**							
	Interval*	0.01	0.01	1.2	0.23	1.01	0.99–1.03	1.4
	LBD†	0.21	0.02	8.5	<0.001	1.23	1.17–1.29	71.9
	Thickness†	0.12	0.03	3.5	<0.001	1.12	1.05–1.20	12.1
	CBI‡	0.13	0.42	0.3	0.76	1.14	0.50–2.61	0.1
	EXE§	1.20	0.83	1.4	0.15	3.31	0.65	2.1
	Multivariate							
	Interval*	0.02	<0.01	4.8	<0.001	1.02	1.01–1.03	23.4
	LBD†	0.21	0.03	8.4	<0.001	1.24	1.18–1.30	71.1
	Thickness†	0.01	0.04	0.1	0.95	1.00	0.93–1.08	<0.1
Enucleation, n = 355	**Univariate**							
	Interval*	0.02	<0.01	4.7	<0.001	1.02	1.01–1.03	22.2
	AJCC stage	0.77	0.10	7.9	<0.001	2.16	1.79–2.62	62.0
	Cytomorphology‖	0.68	0.18	3.8	<0.001	1.97	1.39–2.80	14.5
	BAP-1 expression¶	2.06	0.97	2.1	0.03	7.86	1.17–53.00	4.5
	Multivariate							
	Interval*	0.06	0.01	4.3	<0.001	1.07	1.04–1.10	18.6
	AJCC stage	2.83	0.64	4.4	<0.001	16.9	4.79–59.80	19.3
	Cytomorphology‖	4.03	1.70	2.4	0.02	56.33	2.03–1565.10	5.6
	BAP-1 expression¶	1.72	1.03	1.67	0.09	5.60	0.74–42.10	2.8

AJCC = American Joint Committee on Cancer; CI = confidence interval; CBI = ciliary body involvement; EXE = extraocular extension; LBD = largest basal tumor diameter; SE = standard error.
*Per additional 10-day interval between diagnosis and treatment.
†Per increased millimeter.
‡CBI versus no CBI.
§EXE, yes versus no.
‖Tumor cell cytomorphology, epithelioid versus mixed versus spindle.
¶Nuclear immunoreactivity, low versus high.

There are also several limitations to this study. First, treatment timing—whether prompt or delayed—was not randomized. As a result, the observed differences in outcomes could be influenced by unaccounted variables. For instance, if there was a prevalent, unidentified risk factor among patients who received treatment more than 1 month after diagnosis, this factor might be the real reason for their decreased survival, rather than the treatment delay itself. Although the AJCC stage, BAP-1 immunohistochemical expression, and cytomorphological features are known to hold significant prognostic value, the latter 2 were accessible only for smaller groups of patients who underwent enucleation. Additionally, the interpretation of BAP-1 immunohistochemical staining is subject to variability, because it can be interpreted differently by individual pathologists. The study did not evaluate certain genetic and cytogenetic prognostic markers, including the status of chromosome 3 and gene expression classification. Furthermore, the potential impact of *SF3B1* mutations was not accounted for, known to predispose patients to late metastasis, introducing an element of uncertainty.[40] However, considering those treated promptly had larger tumor LBD, increased thickness, more advanced AJCC stage, and a higher rate of BAP-1 loss in tissue samples, and a similar distribution of factors such as CBI, EXE, patient age, vasculogenic mimicry, and cytomorphology, it is not obvious why the delayed treatment group should have an overrepresentation of such unaddressed adverse prognostic factors.

Second, the nonsignificant associations among CBI, EXE, and metastatic death in univariate competing risks regressions of the subgroup of patients treated with plaque brachytherapy are likely influenced by a selection bias, where tumors with extensive CBI or EXE were more likely to be enucleated.

Third, although it could be assumed that this would influence the results for patients in different groups to a similar extent, causes of death rely on accurate classifications by pathologists, general practitioners, or others responsible for filling out cause of death certificates.

Last, although our data encompass 98% of the initially included patients and 81% of all patients diagnosed with posterior UM in Sweden during the examined period, we cannot discount the possibility that there was an uneven distribution of risk factors among included and excluded patients that could have influenced the results. Although prospective randomized trials would ideally settle the prognostic implications of delayed treatment definitively, the ethical considerations surrounding such a trial would be concerning.

Conclusions

In this cohort, increasing time between posterior UM diagnosis and treatment was associated with increased risk for metastatic death. The association was consistent when evaluating prompt treatment (within 1 month from diagnosis) versus delayed treatment ($\geq$ 1 month from diagnosis) in cumulative incidence analyses of patients in AJCC stage II, stage III, and stage I to III combined; and when using increasing 10-day intervals in competing risks regressions, adjusted for other relevant prognostic factors. Although the link between delayed treatment and adverse prognosis was not statistically significant for patients in AJCC stage I alone, it is notable there was a pronounced surge in UM mortality 10 to 11 years postdiagnosis for those receiving delayed treatment in this group. These results challenge the prevailing theory of early metastatic seeding in UM, emphasize the importance of prompt treatment, and merit further validation in independent cohorts with longer follow-up and comprehensive tumor genetic data.

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Data Availability Statement: Patient-level data for this study are available at https://rcsyd.se. Approval from the Swedish Ethical Review Authority is required before access can be granted.

HUMAN SUBJECTS: Human subjects were included in this study. This study was approved by the Swedish Ethical Review Authority (reference 2022-05357-01). Informed consent was waived because all data utilized were previously recorded in our registries, and no identifiable information such as names, identification numbers, addresses, contact details, or photographs had to be collected. The histological and immunohistochemical data, originally collected for clinical diagnostic and prognostic purposes, were used without conducting additional analyses of biological tissues specifically for this project. All research adhered to the tenets of the Declaration of Helsinki.

No animal subjects were used in this study.

Author Contributions:

Conception and design: Stålhammar

Data collection: Stålhammar

Analysis and interpretation: Stålhammar

Obtained funding: Stålhammar

Overall responsibility: Stålhammar

Abbreviations and Acronyms:

AJCC = American Joint Committee on Cancer; **BCVA** = best-corrected visual acuity; **CBI** = ciliary body involvement; **CI** = confidence interval; **EXE** = extraocular extension; **LBD** = largest basal tumor diameter; **OD** = optic disc; **SD** = standard deviation; **UM** = uveal melanoma.

Keywords:

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